Revitalising Endangered Languages and Cultural Heritage through Language Technology

A Pilot Study for Dzardzongke

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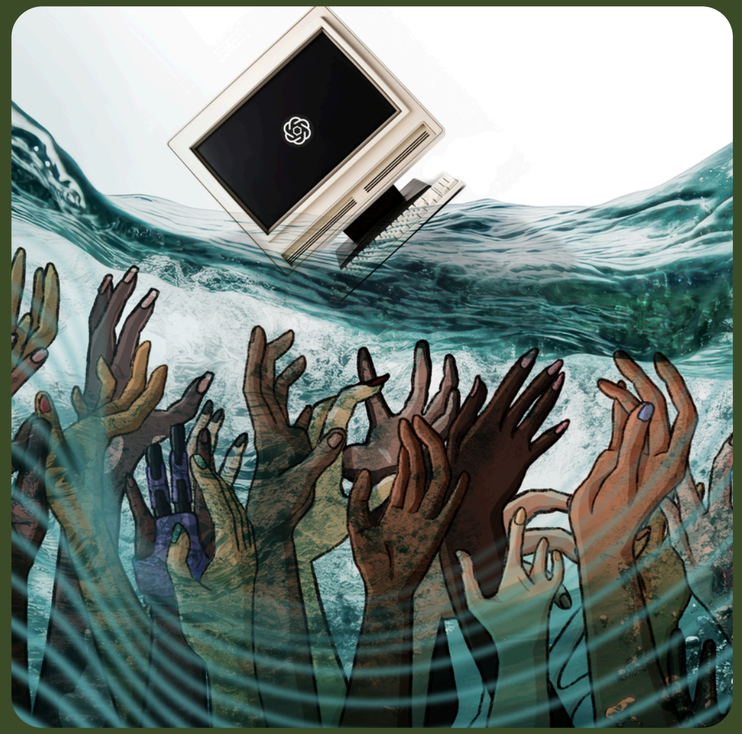
Motivation

Why language technology for endangered languages?

- Thousands of languages remain digitally underrepresented
- Most NLP focuses on high-resource languages
- Digital exclusion accelerates language loss
- Mobile technology offers a scalable opportunity for revitalisation

Goal:

- preserve language
- preserve cultural heritage
- support literacy



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Dzardzongke

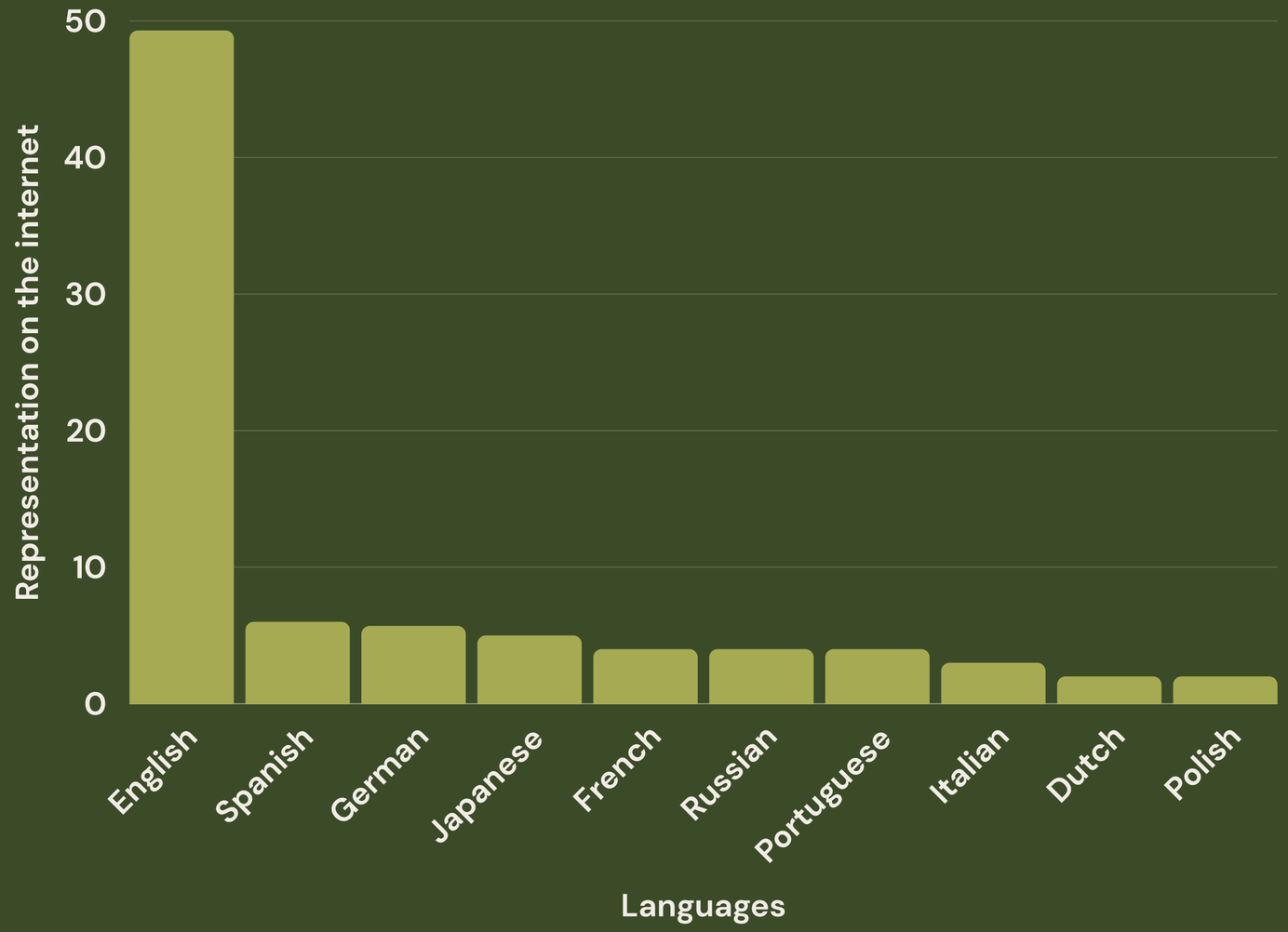
An endangered oral language

- Tibetic language spoken in South Mustang, Nepal
- Approximately 1,200 speakers
- Primarily oral language
- Newly developed Latin-based orthography
- Very limited digital resources
- Increasing smartphone adoption

one of the goals:
not teaching speaking,
but connecting speech to writing

South Mustang

Language Representation on the Internet



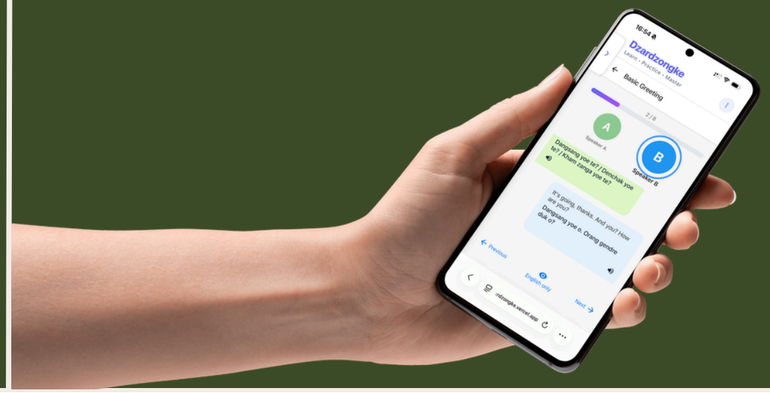
- There are 7,000+ languages worldwide
- Half of the internet's content is in English
- But English is only spoken by 1.5 billion people
- With only 26% being native speakers

Dzardzongke's representation: 0.0%

What are we building?

A community-driven mobile application that supports:

- literacy
- vocabulary learning
- cultural heritage
- long-term language preservation



Key design principles:

- participatory design
- offline-first
- reusable framework
- multimedia learning

Data Collection

- Fieldwork (2022 & 2025)
- Two recorded conversations
- 500+ lexical items
- Audio recordings
- Images

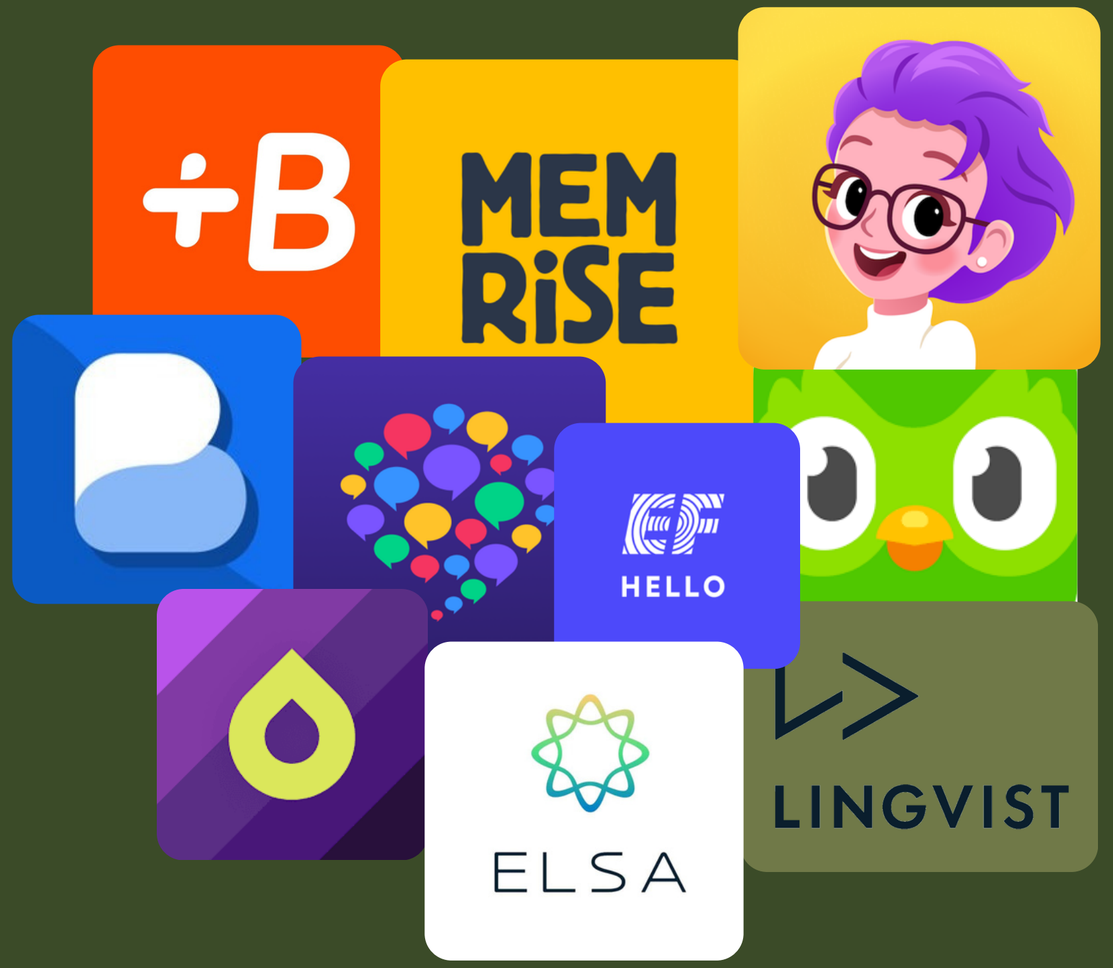
Cultural information:

- villages, information about locations
- festivals
- traditions and customs

Survey of Language Learning Apps

Selected Apps:

- Babbel
- Duolingo
- Memrise
- Busuu
- HelloTalk
- EF Hello
- Lingvist
- EWA
- ELSA Speak
- Drops



Survey of Language Learning Apps

onboarding and user journeys

(e.g. What profile information is collected from users, and how are learning goals set?)

vocabulary presentation and practice mechanisms

(e.g. Which spaced repetition or drilling techniques are used, and how is new vocabulary introduced?)

support for speaking, listening, and literacy

(e.g. How are audio, image, and text combined to support multimodal learning?)

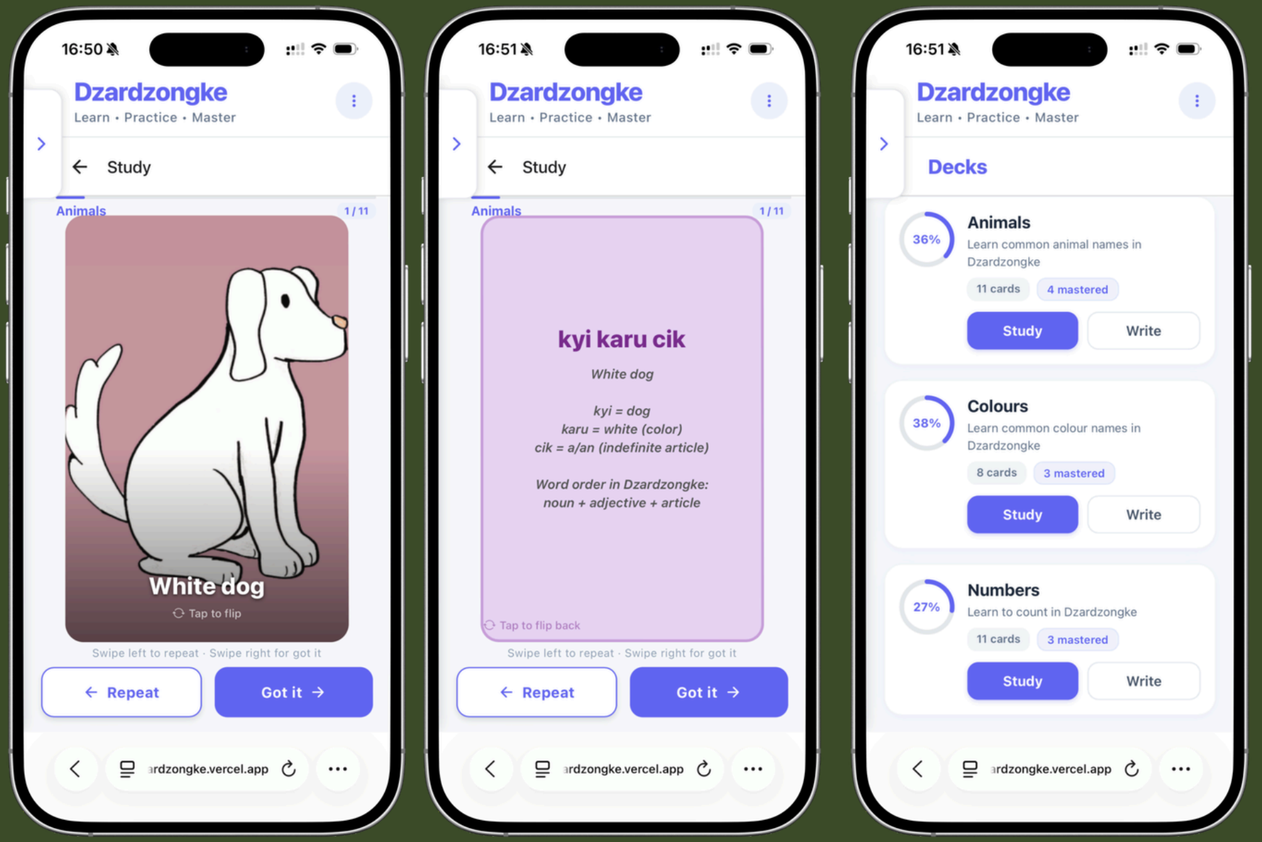
treatment of cultural content and community perspectives

(e.g. Does the app provide cultural context alongside linguistic content, and does it facilitate interaction with other speakers?).

What does our app include?

Main differences:

- users already speak Dzardzongke
- focus on literacy
- simple design

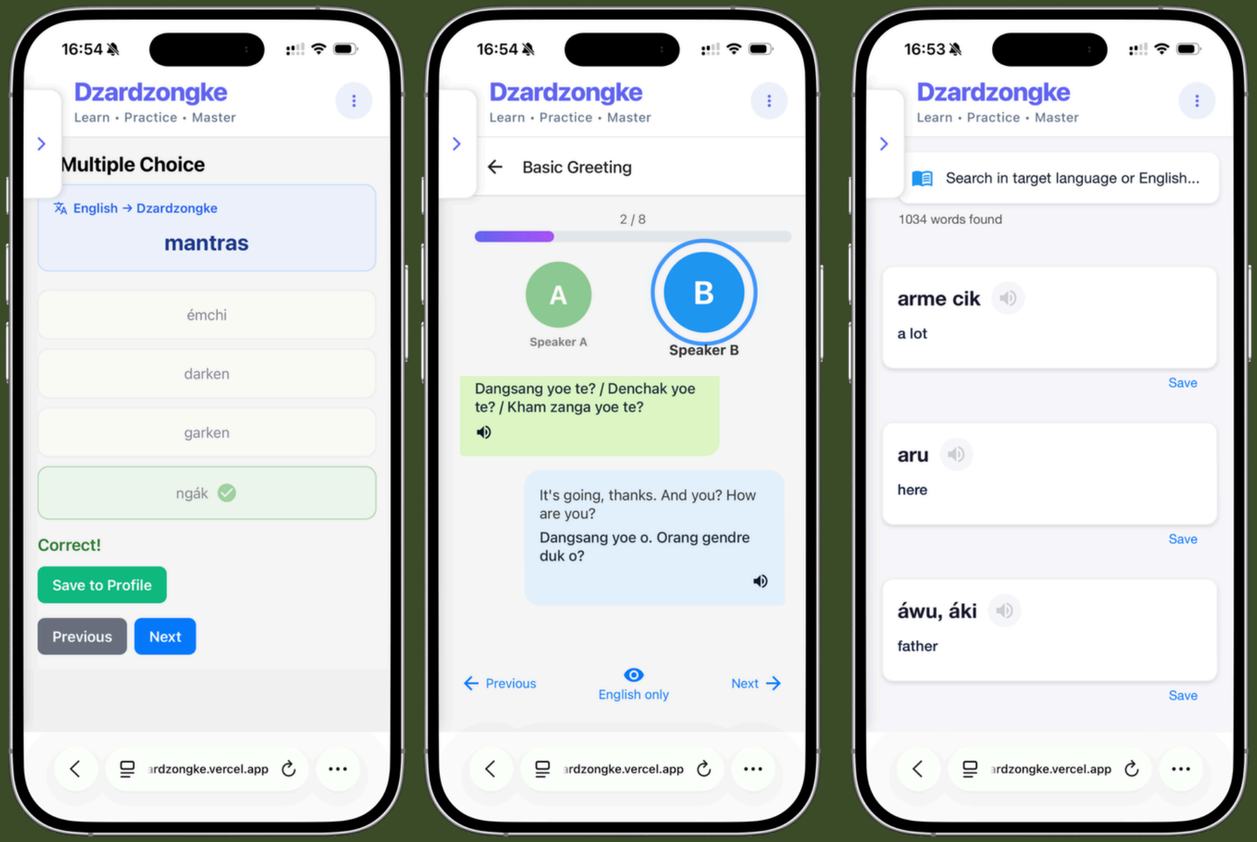


Flashcards - Vocabulary Decks, with progress overviews

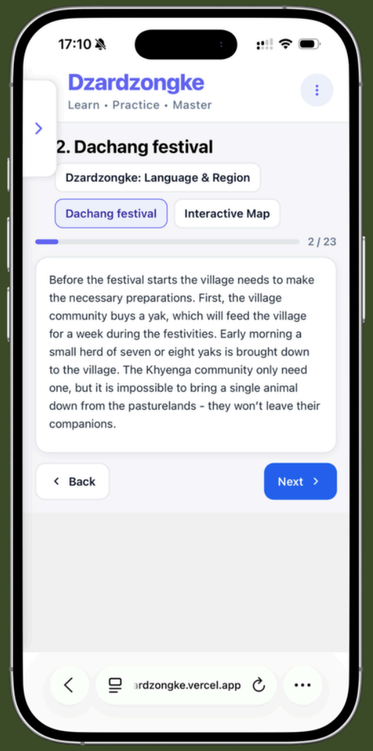
What does our app include?

Additional Features:

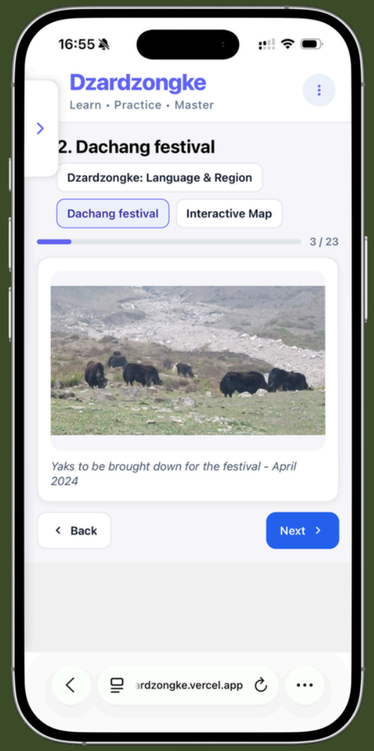
- Quizzes
- Dialogues (recorded with native speakers)
- Dictionary with fuzzy search and connected audio of pronunciation



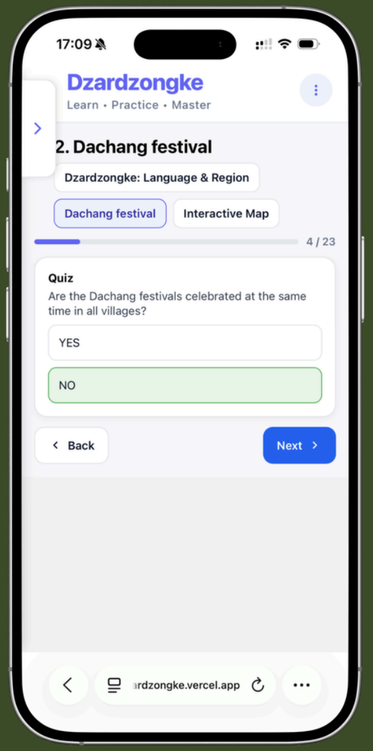
... and most importantly: culture



Cultural information



Images



Tests

Interactive map

Technical Details

- Language Resources:
  - JSON files, audio, images, text
  - All materials archived at ELAR
- React Native App + Expo
- Available on:
  - iOS
  - Android
  - Web

Additional Features:

- offline-first
- local storage
- modular architecture
- language-independent infrastructure

Other researchers/communities can adapt this framework to other oral/endangered languages

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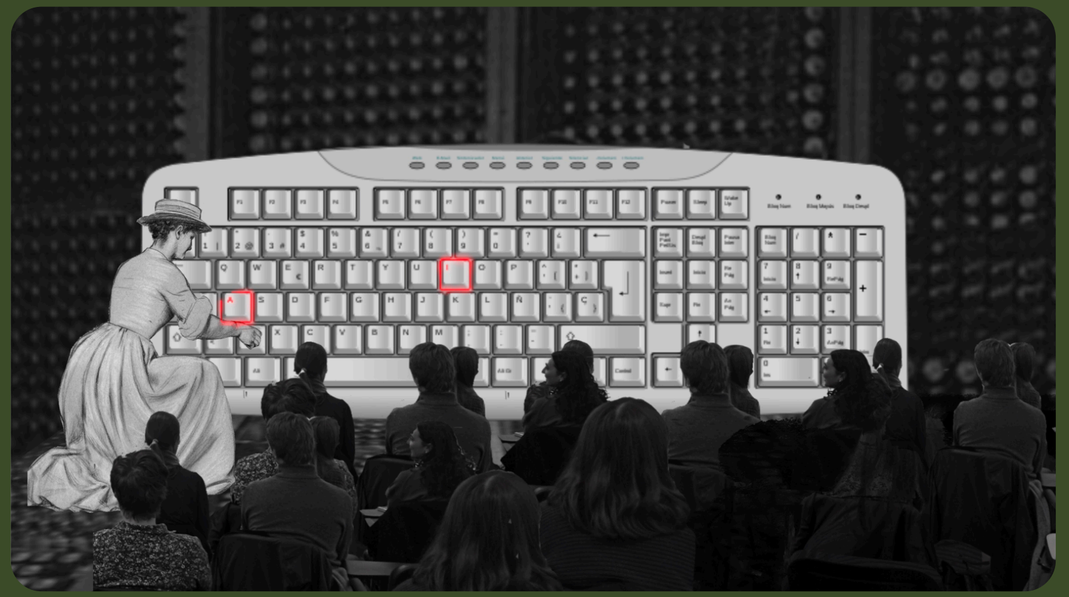
Keyman Keyboard

Supporting writing beyond the app

- Supports the romanised Dzardzongke orthography
- predictive text & autocorrect
- usable inside and outside the app
- enables users to quickly recognise how to spell words they use in their daily lives, even if they have not learnt how to read and/or write them before
- publicly available



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Contributions

- ✓ First mobile app for Dzardzongke + keyboard
- ✓ Integrated language learning + cultural heritage
- ✓ Reusable framework for other endangered languages

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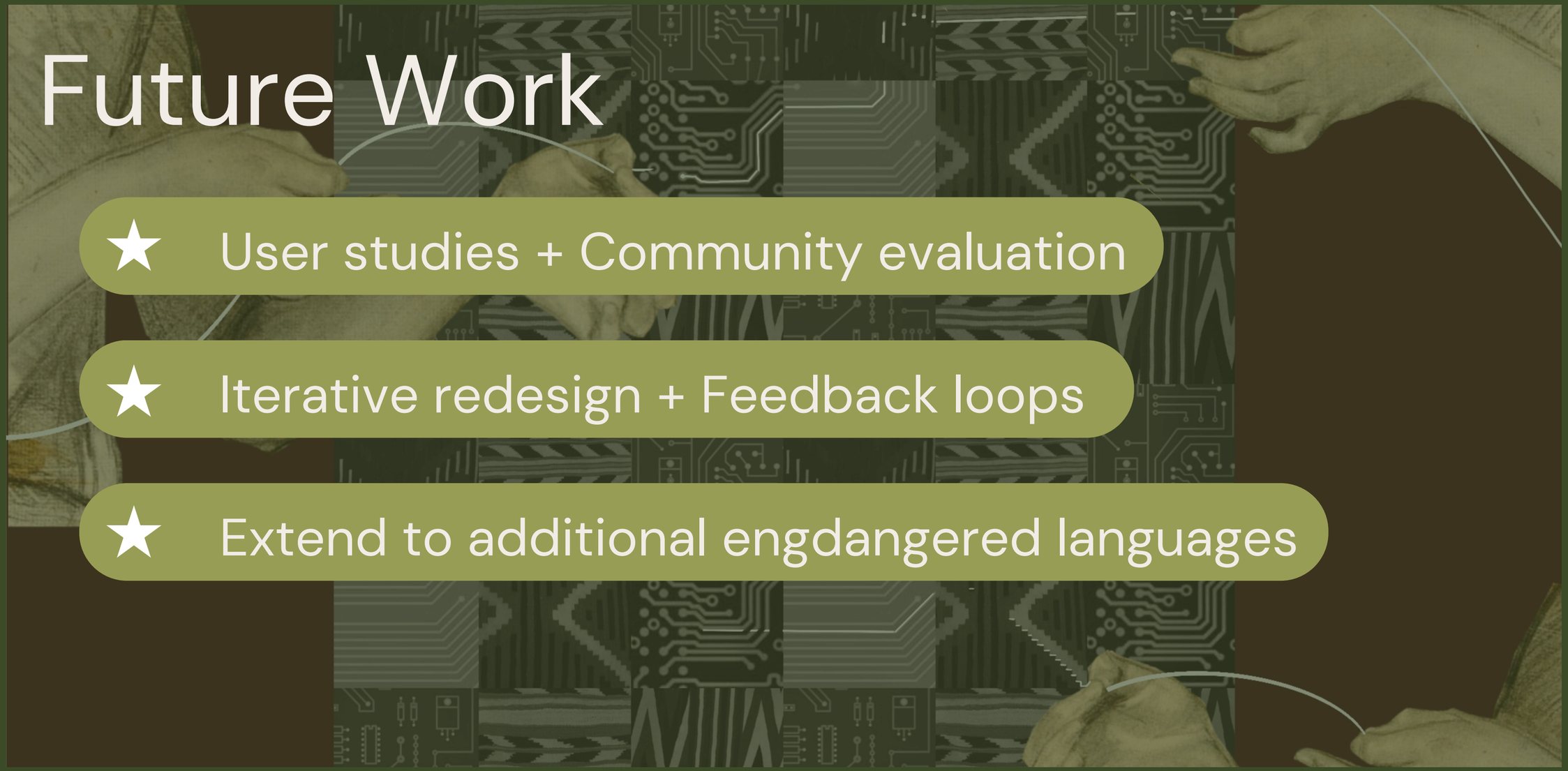
Future Work



- ★ User studies + Community evaluation
- ★ Iterative redesign + Feedback loops
- ★ Extend to additional engdangered languages

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Closing

- Language technology can support endangered language revitalisation
- Community-centred design is essential
- Cultural heritage motivates language preservation
- Our framework can be adapted for many oral/endangered languages

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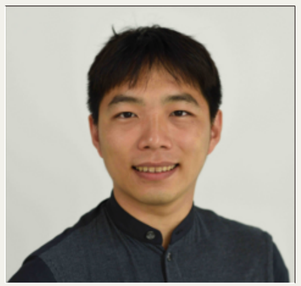

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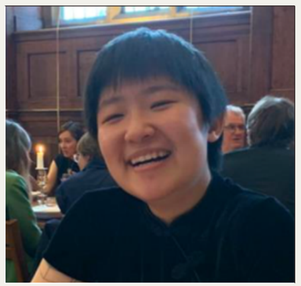

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cambridgeltl/Language-Learning-App-for-Dzardzongke

https://dzardzongke.vercel.app/

Link to the paper:

Thank you!

Q&A